



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2017
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

18 September 2017

2 hours

Candidates answer on the Question Paper.

No additional materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
Section A	
1	/ 24
2	/ 18
3	/ 8
Section B	
4 or 5	/ 25
Total	/ 60

This paper consists of **xx** printed pages.

[Turn over]

Section A

Answer **all** the questions in this section.

QUESTION 1

Some types of snake kill their prey and defend themselves by means of a poisonous bite. Fangs (hollow teeth) inject venom from specialized glands into the victim. The venom contains a protein, which is a toxin.

Different species of snake have toxins that act in different ways. Hemolytic toxins are enzymes that hydrolyze phospholipids. They damage tissues, including heart muscle, which lead to cardiac arrest. Neurotoxins, such as the one produced by green mamba snakes, bind to receptor proteins on the surface membranes of nerve cells or muscle fibers. This interferes with the transmission of nerve impulse, leading to muscle paralysis and heart failures.

Fig. 1.1 shows the molecular structure of fasciculin-2, a neurotoxin produced by the green mamba snake.

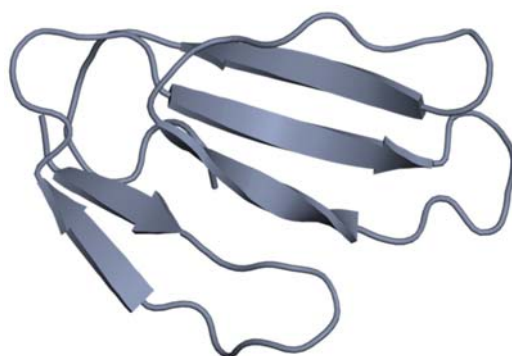


Fig. 1.1

a) Describe the molecular structure of fasciculin-2. [2]

- Composed of only five antiparallel β -pleated sheets as the secondary structures, held by hydrogen bonds between C=O and NH of the peptide region.
- The single polypeptide further folded into a tertiary/globular structure, held by ionic bonds, hydrogen bonds, disulfide bonds and hydrophobic interactions between R groups.

[Award 1m if point 1 and 2 are mentioned without bonds]

b) With reference to hemolytic toxins and neurotoxins, explain why snake venom, which has been heated to 100°C for several minutes, would likely lose its toxicity. [4]

[General effect of heat on proteins]

- Heat breaks hydrogen bonds and ionic bonds that hold the toxin in its intact 3D structure
- Loses tertiary / globular / 3D structure, hence denatures the toxin

[Effect of heat on hemolytic toxins]

- Hemolytic enzymes loses the shape of its active site, hence no longer complementary in shape [mark once] to phospholipids, hence unable to bind and hydrolyze phospholipids

[Effect of heat on neurotoxins]

- Shape of neurotoxin is no longer complementary to the receptors on nerve/muscle cells, hence will not interfere with nerve impulse transmission.

c) State how enzymes which hydrolyze phospholipids damage tissues. [1]

- Disrupts cell membrane integrity leading to lysis of cells.

d) Some antibodies bind to toxins and inactivate them. These antibodies are known as anti-toxins. The human immune response is far too slow to be effective in making anti-toxins against snake venom.

Injecting a very small, non-lethal quantity of venom into a horse produces anti-toxin. The horse produces anti-toxins that can be extracted from horse blood and used as an emergency treatment for those bitten by the same species of snake. Each time the horse is injected with venom, it is able to tolerate larger doses and the concentration of the specific anti-toxin in its blood is greater.

i) Explain why the human immune response is too slow to protect a person from a snake bite. [3]

- Toxin acts too fast for antitoxin to develop
- Time is needed to [max 2]
 - o Internalization of toxin by B cells and macrophages
 - o Antigen presentation in complex with MHC class II by B cells and macrophages to helper T cells
 - o Proliferation and differentiation of B cells into plasma cells
 - o Antibody synthesis and secretion from plasma cells
- Human unlikely to have been bitten before, hence no prior exposure to the toxin, thus has no memory B and T cells to immediately respond to the toxin

ii) Explain why a horse is injected more than once with a small amount of venom when it is being prepared for use as a source of anti-toxin. [2]

- First injection stimulates a primary immune response to produce anti-toxin-specific memory B cells and plasma cells, where only a small amount of anti-toxin are produced.
- Second injection stimulates secondary immune response, where many memory B cells differentiate into many plasma cells, hence a large amount of anti-toxin is secreted.
- **AVP** e.g. a single large dose would kill the horse

iii) It was observed that upon injection with the toxin, different types of anti-toxins are produced. Each type of anti-toxin is different at the variable region, but they are equally effective against the toxin.

Suggest why different anti-toxins are produced. [2]

- Toxin is an antigen with different epitopes.
- Each (naïve) B cell only recognize a single epitope, thus, each B cell differentiate into plasma cells that produce antitoxin specific to that epitope

iv) Explain why treatment with the horse anti-toxin will not produce long-term protection against snake bites. [3]

- **Ref to.** Passive immunity
- Anti-toxin remains in the blood only for a short time / eventually destroyed
- The person's immune system is not stimulated to produce anti-toxin, hence no memory B cells produced
- **AVP** e.g. different snakes have different toxins

e) In recent years, contortrostatin, a protein found in the venom of the southern copperhead snake, has been extensively demonstrated to hold great promises in cancer treatment. Contortrostatin binds to and disrupt the function of integrins, causing tissue damage. Integrins are transmembrane proteins that serve as bridges for cell-to-cell interactions, which is important in adhering endothelial cells of blood vessels to our body tissues.

Using the information above and your knowledge on the characteristics of cancer cells, suggest why contortrostatin can be used as a medicine in cancer treatment. [3]

- Prevents angiogenesis / formation or attachment of new blood vessels in a tumour...
- ...hence cuts off the delivery of oxygen and nutrients to cells of the tumour, preventing growth
- **Idea that** Since these vessels provide the principal route by which cancer cells exit the primary tumour and enter the circulation, metastasis, the spread of cancer cells, is hence prevented.

- f) Mice and monkeys have been successfully immunised against several important infectious diseases using experimental DNA vaccines, in the form of plasmids. Plasmids are small circular DNA molecules.

During the 1990s, researchers found that mouse muscle and other mouse tissues were able to absorb plasmids which had been injected into the animals. Any genes that were part of this plasmid DNA were transcribed and translated. The resulting polypeptides were presented on the cell surface in complex with host receptor molecules, which allow the immune system to recognise the polypeptide as non-self. Proteins that are presented at the cell surface in this way stimulate the lymphocytes of the immune system very effectively.

This discovery allows plasmid DNA to be used as a vaccine, even though the DNA does not itself act as an antigen. Most vaccines contain proteins, or fragments of proteins, that are extracted from the surface of pathogens. It is a complex and costly procedure to purify these protein antigens.

Fig. 1.2 shows a simplified diagram of a DNA vaccine. This plasmid codes for two antigens, **A** and **B**.

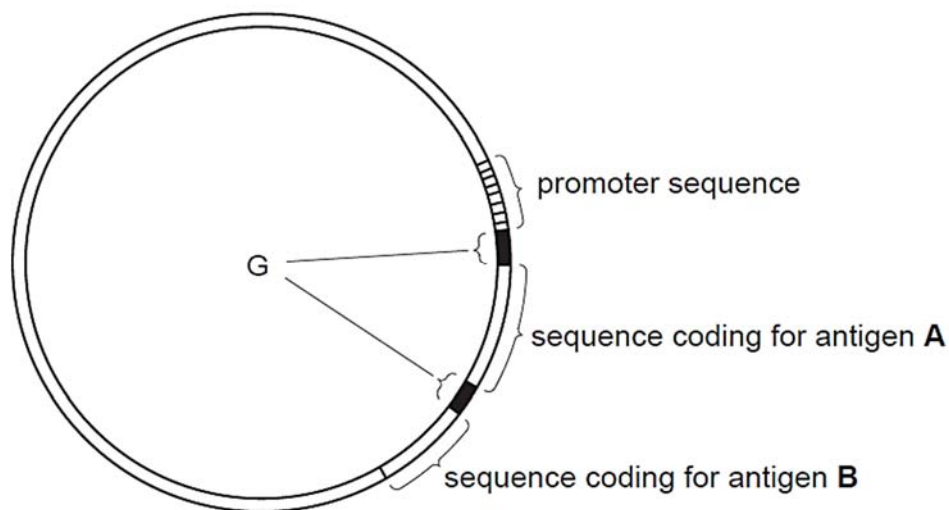


Fig. 1.2

- i) Suggest why proteins presented at the cell surface of antigen-presenting cells are able to stimulate an immune response more effectively than proteins dissolved or suspended in the blood or tissue fluids. [1]

- Proteins presented on the cell surface are able to remain in the body for a long period of time, compared to antigens in blood, which could be rapidly degraded
- Proteins presented on cell surface is able to activate T cells, leading to cell-mediated immunity, but free proteins is unable to do so.

- ii) Sequences of nucleotides, labelled **G** on Fig. 1.2, code for groups of amino acids at the beginning of each polypeptide. These amino acid sequences direct the newly-synthesised polypeptides to the rough endoplasmic reticulum of the muscle cell.

Suggest how this makes the vaccine effective. [2]

- The polypeptide is able to be loaded onto the major histocompatibility complex molecules embedded in the RER membrane
- The polypeptide-MHC complex can then be packaged into ER vesicles to be transported to the cell surface for presentation.

- iii) Suggest why it may be advantageous to include nucleotide sequences coding for more than one antigen in a DNA vaccine. [1]

- **Idea that** protection against more than one pathogen
- **Idea that** activate more than one B cells that recognize more than one antigen
- **Idea that** reduces the number of vaccination needed
- **AVP**

[Total: 24]

QUESTION 2

In an attempt to control the spread of dengue, using genetic modification, a piece of DNA is inserted into the *Aedes aegypti* mosquito genome at the embryonic stage. This DNA contains a lethal gene (*tTAV* gene) which codes for a protein called tTAV. This protein acts as a molecular switch to shut down the expression of **all** other genes, leading to death of the insect. In order to shut down all other genes, the tTAV protein concentration in the cell must reach a high concentration. To achieve that, the tTAV proteins that are initially synthesized also function to increase the expression of its own gene, thereby producing even more tTAV proteins, demonstrating what is known as a positive feedback.

This tTAV protein, however, is inactivated by a compound called tetracycline, which is incorporated into the food that the developing larvae feed on. Hence, the genetically-modified (GM) larvae survive to adulthood, with much of the tetracycline still remaining in them. Male GM mosquitoes are then selected to breed with females to produce large number of offspring. The male GM offspring are selected and fed with tetracycline until they reach adulthood. They are then released into the wild to mate with wild-type females. Any offspring larvae produced will contain the *tTAV* gene, which is expressed to cause death of the larvae.

- a) Explain how tTAV protein increases the expression of its own gene. [2]
- Functions as an activator that binds to enhancer region of tTAV gene
 - Subsequent bending of DNA brings the enhancer close to the promoter to stabilize the transcription initiation complex, which upregulate transcription.
- b) Suggest a reason why a high concentration of tTAV proteins would shut down the expression of **all** other genes. [2]
- tTAV proteins binds to transcription factors / RNA polymerase to make them unavailable for transcription.
 - Because there are high amount of transcription factors / RNA polymerase, high concentration is needed.
- OR
- tTAV proteins binds to promoter to block the binding of transcription factors...
 - Because there are many genes, high concentration is needed.
- c) Suggest **two** advantages of using GM mosquitoes over the use of pesticides in controlling the spread of dengue. [2]
- **Idea that** Highly species-specific: the released male GM mosquitoes only mate with the females of their own species. This means that no other insects are affected, unlike the use of pesticides that affects all insects
 - **Idea that** Male GM mosquitoes does not pose any harmful environmental effects, unlike the chemicals found in pesticides.
 - **Idea that** Mosquitoes can develop resistance against pesticide
 - **AVP**

Male GM mosquitoes have been used in open field trials in countries such as Cayman Islands. The town where the *Aedes aegypti* mosquitoes predominate was divided into three areas, as shown in Fig. 2.1.

Area **A** – the treatment site where male GM mosquitoes are released

Area **B** – buffer zone

Area **C** – the non-treated control site

The mosquito populations in area **A** and area **C** were measured using an ovitrap – a device that is attractive as an egg-laying site for female mosquitoes.

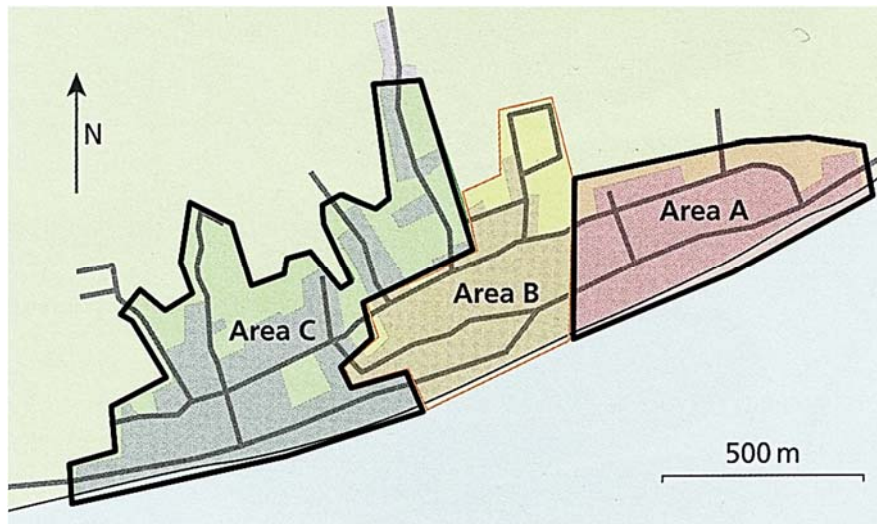


Fig. 2.1

d) State why the release of male GM mosquitoes in area **A** will not increase the risk of transmission of dengue. [1]

- **Idea that** Only female mosquitoes feed on human blood to transmit the dengue virus.

e) **[HI-1]** Suggest the purpose of area **B**. [1]

- **Idea that** Ensure GM mosquitoes were unlikely to fly to non-treated control site, which could interfere with the results.

The number of ovitraps that contain eggs were recorded every week for 6 months. Fig. 2.2 shows the ovitrap index, which is calculated based on the percentage of ovitraps containing eggs.

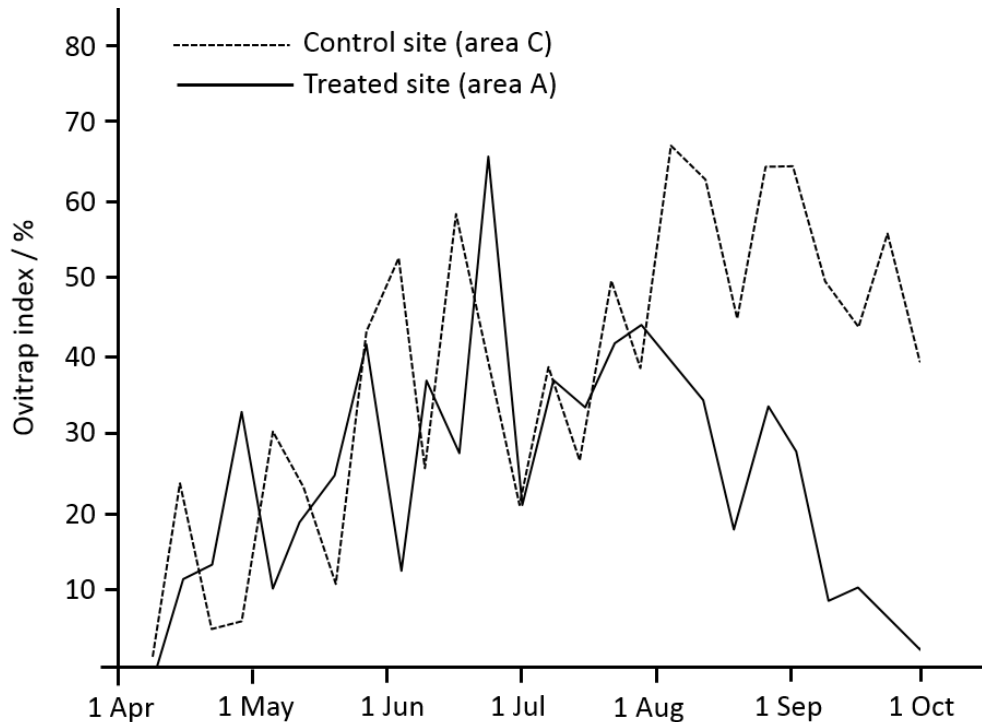


Fig. 2.2

f) Comment on the trend observed for both the treated and control site.

[4]

- For both sites, ovitrap index fluctuates from Apr to Oct.
- Rises due to females laying eggs, falls due to the hatching of the eggs
- For control site, upward trend from Apr to Oct, but for treated site, upward trend from Apr to mid-Jun to Aug, then downward trend till Oct.
- By 1 Oct, ovitrap index stands at 40% at control site, but only 2% at treated site
- Male GM mosquitoes pass on the tTAV gene to offspring, causing the larvae to die before reaching adulthood, hence less adult mosquitoes mate to lay eggs.

- g) Global warming is the unusually rapid increase in Earth's average surface temperature over the past century primarily due to the greenhouse gases released as a result of anthropogenic activities. The global average surface temperature rose by 0.6 to 0.9°C between 1906 and 2005, and the rate of temperature increase has nearly doubled in the last 50 years. Temperatures are certain to go up further.

i) Explain how global warming has encouraged the spread of dengue. [3]

[Effect of temperature]

- Higher temperature hasten the life cycle of mosquitoes due to increased metabolism, hence producing more offspring
- Higher temperature causes female mosquitoes to feed more frequently due to increased rate of digestion, this increases transmission intensity
- Temperate countries are now experiencing a warmer temperature, thus encouraging mosquitoes to migrate to higher latitudes

[Effect of precipitation]

- Global warming has also caused increased precipitation, hence increases the number of breeding sites for mosquitoes.

- ii) Apart from the spread of mosquito-borne diseases, global warming is already putting pressure on ecosystems (the plants and animals that co-exist in a particular climate zone), both on land and in the ocean. Warmer temperatures have already shifted the growing season in many parts of the globe. Spring is arriving earlier in both hemispheres, causing the growing season in parts of the Northern Hemisphere becoming two weeks longer in the second half of the 20th century.

This change in the growing season also affects the broader ecosystem. Migrating animals have to start seeking food sources earlier. Furthermore, the shift in seasons may already be causing the life cycles of pollinators, like bees, to be out of sync with flowering plants and trees. This mismatch can limit the ability of both pollinators and plants to survive and reproduce, which would reduce food availability throughout the food chain.

Describe how global warming has impacted **other** biotic factors. [3]

[Death of coral reefs]

- Warmer sea, more CO₂ dissolved, more acidic, dissolves away CaCO₃ exoskeleton of coral reefs.
- Warmer sea, coral reefs expel photosynthetic zooxanthellae, leading to coral bleaching

[Threat to global food supply]

- **Ref to.** livestock death due to heatwaves / increased prevalence of parasites and diseases / decreased forage area due to drought / AVP
- **Ref to.** drop in fisheries due to lesser fishes in warmer water / competition of local species with invading species / AVP

[Reduction in rich biodiversity]

- **Ref to.** loss of genetic diversity for food
- **Ref to.** loss of biomedicines – some plants contain bioactive compounds that can be used for pharmaceutical purposes
- **Ref to.** the loss of habitats leading to extinction of certain species

[Total: 18]

QUESTION 3

Ribulose biphosphate carboxylase-oxygenase, better known as rubisco, is a massive protein made up of 16 subunits, and is an important enzyme that all life forms depend on. It has a low affinity for carbon dioxide and fixes only 3–10 molecules of carbon dioxide per second, compared to other enzymes which convert hundreds to millions of substrates per second. The consequence is that photosynthetic cells synthesize a large amount of rubisco. About half of all proteins in green leaves consist of rubisco, making this enzyme the world's most abundant protein.

In addition to carbon dioxide, the same active site of rubisco that binds carbon dioxide also binds oxygen, hence this enzyme has 'oxygenase' in its name. The oxygenase activity of rubisco combines oxygen to RuBP, which is split into 3-phosphoglycerate and a two-carbon compound called 2-phosphoglycolate, as shown in Fig. 3.1.

2-phosphoglycolate is converted into 3-phosphoglycerate in a series of reactions, but this pathway consumes oxygen and releases carbon dioxide, hence the name *photorespiration* is given to this pathway. The rate of photorespiration is usually about one-third that of the Calvin cycle, but this rate is predicted to increase with global warming, reducing plant productivity.

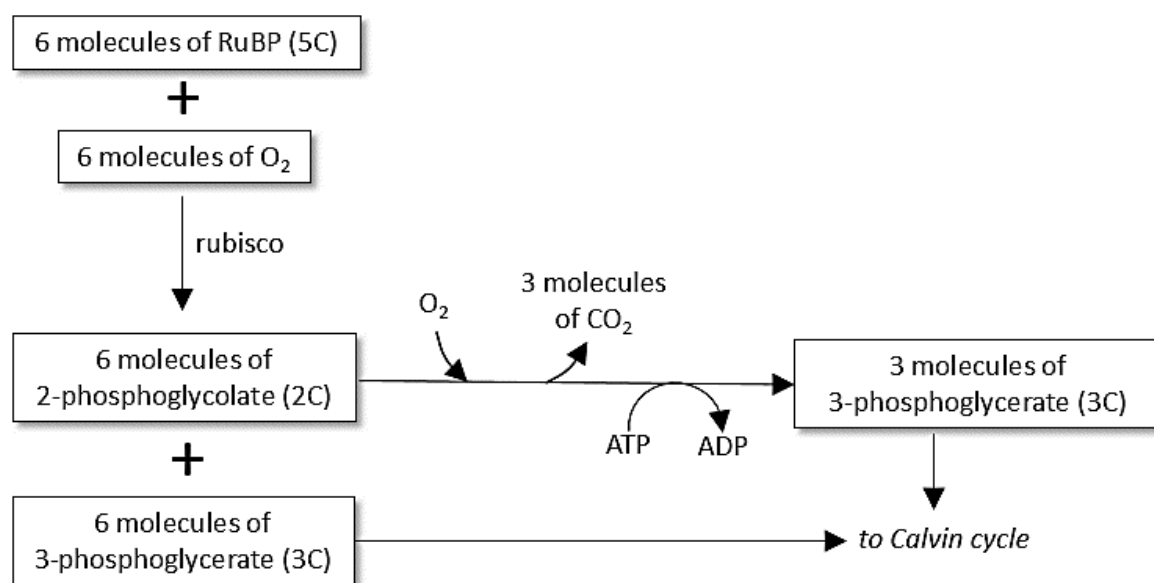


Fig. 3.1

a) Explain why all life forms are dependent on rubisco.

[3]

- First enzyme in Calvin cycle that combines carbon dioxide with RuBP / carbon fixation to form two molecules of 3-phosphoglycerate.
- Calvin cycle synthesizes glyceraldehyde-3-phosphate which is then used to make glucose.
- **Idea that** glucose is used by the plant itself for growth as well as by primary consumers and subsequent consumers (for respiration).

b) Explain why photorespiration will reduce the rate of Calvin cycle.

[2]

- Photorespiration uses ATP...
- ...which deprives Calvin cycle of it in the reaction, where 3-phosphoglycerate is converted to 1,3-bisphosphoglycerate / needed for RuBP regeneration.

c) Rubisco has been described as a classic example of '*unintelligent evolutionary design*'.

Explain why.

[3]

- Low affinity for carbon dioxide (substrate) as compared to other enzymes
- Low catalysis rate of 3-10 carbon dioxide per second as compared to other enzymes which catalyzes thousands to millions of substrates per second.
- **Idea that** Deprives itself of binding to carbon dioxide by also binding to oxygen, which produces 2-phosphoglycolate and 3-phosphoglycerate.
- **Idea that** When 2-phosphoglycolate is converted to 3-phosphoglycerate, carbon dioxide is lost in photorespiration, rather than being fixed.

[Total: 8]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 4

- a) DNA molecules replicate with a high degree of accuracy, yet not always perfectly.

Describe how this occurs and discuss why the survival of a species depends on DNA molecules being stable, yet not *absolutely* stable. [13]

[How DNA replication takes place accurately] – max 3

1. DNA is double-stranded, each strand is complementary to the other
2. Each strand acts as the template for synthesis of daughter strand by complementary base pairing (A=T, C=G)
3. DNA polymerase III with proofreading function / 3'→5' exonuclease activity
4. Able to excise previous nucleotide that is wrongly paired and replace with the correct nucleotide
5. DNA polymerase I with proofreads newly-synthesized daughter strand / 5'→3' exonuclease activity

[Why DNA replication is not always perfect] – max 3

6. Exposure to radiation / chemical carcinogens / **AVP**
7. Causes structural damage to DNA + *cite an example below*
 - o e.g. UV light causes thymine dimer formation
 - o e.g. chemicals (such as nitrous acid) chemically reacts with base
 - o e.g. ethidium bromide intercalates into DNA
8. Such structural damage causes wrong nucleotide(s) / extra nucleotide(s) / missing nucleotide(s) to be added during DNA replication.
9. Spontaneous mutation – DNA polymerase adds the wrong base, and is not being rectified.

[Why survival of offspring depends on DNA being stable] – max 3

10. **Idea that** Ensures sequence of DNA in genes is intact so that (normal amount of) functional proteins can be made
11. **Idea that** Mutation results in non-functional / hyperactive / overproduction / underproduction of proteins
12. **Ref to** Sickle-cell anemia – Single-base substitution to β -globin gene that causes Hb to crystallize, forming sickle-cell RBC which clogs blood vessels / inefficient O₂ transport
13. **Ref to** Cancer – a result of gain-of-function mutation to proto-oncogenes and loss-of-function mutation tumor-suppressor genes, leading to uncontrolled cell division.

[Why survival also depends on DNA being not *absolutely* stable] – max 4

14. **Ref. to** role of mutation in natural selection
 - a. Mutations allow for formation for new alleles
 - b. Provides variation between individuals in a population to allow the population to respond to environmental change
 - c. Survival of the fittest to allow population to evolve, hence prevents extinction of a species
15. **Ref. to** [for mammals] role of somatic recombination of antibody genes in B cells (and T cells)
 - a. Generate millions of different B cells, each with a B-cell receptor specific for an antigen
 - b. Ready to combat against any pathogen to prevent damage to body
16. **Ref. to** [for mammals] role of somatic hyper-mutation during affinity maturation of B cells
 - a. High rate of mutation to the rearranged V(D)J segment after activation by an antigen
 - b. Results in B cells that possess increased affinity to the antigen

QWC: Argue that continuation of a species and its continued evolution relies on a balance between accurate transmission of nucleotide sequences and the need for random change to provide the variation needed to allow continued evolution, thus responding to environmental change

b) Discuss the view that all life forms depends on phosphate. [12]

[Nucleic acids]

1. Phosphate being one of the component of a nucleotide
2. Needed to form phosphodiester bonds in a polynucleotide
3. DNA: contains genetic information needed to synthesize proteins for cells to function
4. mRNA: conveys genetic information from nucleus to the cytoplasm
5. tRNA: carries amino acids to the ribosome for synthesis of polypeptide
6. rRNA: forms part of ribosome, the translation machinery
7. telomerase RNA: forms part of telomerase, where it is a template for extension of telomere
8. snRNA: part of spliceosome, needed for RNA splicing to produce mature mRNA

[Phospholipids in biological membranes]

9. Forms phospholipids, which is the building blocks of biological membranes
10. Due to its hydrophilicity, membrane forms a bilayer, where the phosphate group faces the aqueous external environment and aqueous cytosol
11. Membranes are fluid, which is important for substances to be transported in and out of cell
12. Phosphate of phospholipids also interact with proteins to allow their embedment
13. Phosphate of phospholipids also interact with cholesterol to regulate membrane fluidity

[ATP]

14. Energy molecule that releases energy upon hydrolysis of phosphate bond
15. For phosphorylation of glucose and fructose during glycolysis
16. To convert glycerate-3-phosphate to 1,3-bisphosphoglycerate in Calvin cycle
17. For active transport of substances against concentration gradient
18. Named example: e.g. pump protons from cytosol into lysosomes to maintain acidic pH
19. For movement of vesicles within the cell
20. As a substrate for adenyl cyclase to produce the second messenger cyclic AMP (cAMP)
21. **AVP**

[GTP]

22. To activate G-protein

[Protein activation via phosphorylation by kinases]

23. Needed by kinases to phosphorylate and hence activate proteins e.g. during phosphorylation cascade in signal transduction

[Total: 25]

QUESTION 5

- a) It is observed that cancer cells share similar characteristics as stem cells, yet there are characteristics which distinguish them.

Describe the above observations and explain the molecular basis of cancer. [13]

[Similar characteristics] – max 2

1. Unspecialized/undifferentiated – lack of tissue-specific structures which allows them to perform specialized function.
2. Capable of dividing for an indefinite period of time
3. Telomerase genes are expressed – high telomerase activity

[Distinguishing characteristics] – max 2

4. Controlled vs uncontrolled cell division
5. Stem cells can differentiate into specialized cells, but cancer cells remain undifferentiated.
6. **Ref to.** loss of cell cycle checkpoints in cancer cells, but checkpoints intact in stem cells.

[Molecular basis of cancer] – max 9

7. Gain-of-function mutation to proto-oncogenes to become oncogene
 - a) Normal proto-oncogene codes for proteins that drive normal cell division
 - b) Oncogene codes for hyperactive proteins / excessive amount of normal proteins, leading to over-stimulation of the cell cycle
 - c) Oncogene is a dominant allele – one copy is sufficient to induce cancer formation
 - d) **Example:** ras gene that codes for a G-protein that leads to the expression of proteins that stimulate the cell cycle
8. Loss-of-function mutation to tumor-suppressor genes
 - a) Normal TSG codes for proteins that inhibit uncontrolled cell division
 - b) Mutated TSG codes for proteins that are nonfunctional or in reduced amount
 - c) Mutated TSG is a recessive allele – both copies must be mutated to induce cancer formation
 - d) **Example:** p53 gene codes for the an activator that initiates transcription of genes involved in apoptosis / halting of cell cycle / repair of DNA.
9. Multi-step process – a single cell needs to accumulate mutations in both tumour suppressor genes and proto-oncogenes to become cancerous.
10. **Ref. to** causative factors that increases chances of DNA mutations (e.g. carcinogens, radiation, infection by some viruses).
11. **Ref. to** genetic predisposition to cancer if inherit one copy of oncogene or mutated TSG from parents
12. **Ref. to** angiogenesis and metastasis

b) Discuss the importance of hydrogen bonding in ensuring the continuity of life.

[12]

[Role of H-bonds between complementary base pairs]

1. Allows complementary base pairing to occur in nucleic acid interactions

[DNA]

2. Stabilizes two DNA strands to form double helical DNA molecule
3. **Ref. to** role of DNA (e.g. storing genetic information)

[tRNA]

4. Intra-molecular hydrogen bonding in tRNA allows tRNA to fold into a clover-leaf structure
5. **Ref. to** role of tRNA – carries amino acids to the ribosome for synthesis of polypeptide

[rRNA]

6. Intra-molecular hydrogen bonding in rRNA allows rRNA to fold into a precise 3D structure to complex with ribosomal proteins to form ribosome
7. **Ref. to** role of ribosome – translation machinery

[snRNA]

8. Intra-molecular hydrogen bonding in snRNA allows snRNA to fold into a precise 3D structure to complex with spliceosomal proteins to form spliceosome
9. **Ref. to** role of spliceosome – splicing of primary mRNA transcript to produce mature mRNA

[Telomerase RNA]

10. Intra-molecular hydrogen bonding in telomerase RNA allows telomerase RNA to fold into a precise 3D structure to complex with TERT to form the telomerase enzyme
11. **Ref. to** role of telomerase – restore telomere length to ensure infinite division in stem cells

[During DNA replication]

12. Important in DNA replication, where daughter DNA strand is synthesized via adding complementary deoxyribonucleotides to template DNA to ensure accurate transmission of genetic information.

[During transcription]

13. Important in transcription, where RNA is synthesized via adding complementary ribonucleotides to template DNA

[During translation]

14. Important in translation, where codons on mRNA complementary base pair with anticodon on tRNA to ensure correct sequence of amino acids forms the polypeptide

[Role in maintaining protein structure]

15. **Ref. to** maintaining secondary structures (α -helices and β -pleated sheets) in proteins, formed between peptide regions.
16. **Ref. to** maintaining tertiary/quaternary structure of proteins, formed between R groups.
17. **Idea that** Shape of proteins dictates their specific functions (e.g. in DNA replication and gene expression)

[Role in enzyme-substrate interaction]

18. **Ref. to** allow substrate to bind weakly to the active site of enzyme

[Role in solubility]

19. **Ref. to** allows hydrophilic substances to be soluble in aqueous environment to allow reaction to take place

20. AVP

[Total: 25]